

MAKAR BADERKO

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ABOUT ME

As a passionate 17-year-old Software Engineer, I leverage my interdisciplinary expertise in computer science, economics, and physics to bridge diverse perspectives and ideas of my colleagues in order to foster collaborative synergy and drive innovative solutions for impactful outcomes.

SKILLS

Technical Skills Python, C++, SQL, Classic ML, Computer Vision, Gradient Boosting, Deep Learning with PyTorch, Data Mining, LLM fine-tuning and Stochastic Optimization
Languages Russian - mother tongue, English - C1 (Cambridge certificate), German - B2, Italian - A1
Interests Finance, foreign languages, reading and photography

RESEARCH

- AdLoCo: Adaptive batching significantly improves communications efficiency and convergence for Large Language Models - *ICOMP 2025*

ACADEMIC ACHIEVEMENTS

Computer Science and AI

- National Statistics and Data Analysis Olympiad - Gold(2022, 2024, 2025), Silver(2023)
Top-1 result in the theoretical section (2022, 2024) & all-time national cumulative score record holder
This is the most significant olympiad in data analysis in Russia
- National Technological Olympiad in Artificial Intelligence - Gold(2025)
Top-1 result (among my grade participants)
This is the most significant olympiad in AI in Russia
- Kaggle Competitions Expert - world's top 510 Data Scientist (1x Gold, 1x Silver)

Economics and Business

- WEC Continental Economics Cup - 2nd place in Europe in Business Case Analysis section (scored 190/200) and a Bronze medal overall (2024)
- International Economics Olympiad (Open Track) - 95% in Finance and 98.9% in Economics (top 2 out of 500 in economics in the world)
- Moscow State University Championship in Economics - PwC Russia nomination awardee (2025)

Eligible for merit-based admission to Russia's top-tier universities (MIPT, HSE, MSU, ITMO, BMSTU)

WORK EXPERIENCE

Biometriclabs

Backend Developer

(July 2023 - August 2023)

- Created a high-performance codebase to connect the mathematical models written in C with the existing UI and the database of reports from sensors

Biometriclabs

Embedded Developer

(June 2021 - August 2021)

- Built prototypes of innovative products with the Raspberry Pi, collected human heart rate data and converted it into the required digital format

Granted direct admission to internships in ML at largest russian banks (Sber and Tbank) for winning top corporate ML competitions (interview waived).

PROJECTS

Computer Science and AI

- Machine learning model, that can use publicly available blockchain network data to prevent a transaction with a high risk of money laundering (GNNs and more)
- Amur tiger detection, segmentation and reidentification software for population size estimation
- Unwanted plants dataset collection, detection and segmentation software for farmers

Economics and Business

- Company reviews reliability evaluation for credit scoring using advanced mathematical modelling techniques using data about 15,000+ companies
- Governmental investment effectiveness evaluation and prediction based on overall economy performance using data about 200,000 companies
- Private investment regularity evaluation and prediction for brokerage recommender systems using data about 5000 clients